

Four steering conventions for temporal dictionaries

what each one actually writes into the residual stream

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Abstract

Four procedures in this codebase are all called “steering a temporal dictionary latent”, and they write different things. Two of them — the paper’s `v7` tiled-broadcast hook and this sprint’s `txc_flat` control — turn out to be the *same* intervention up to a scalar, even though one is the published method and the other was built to be a null. A third, the paper’s `pp` per-position hook, collapses onto the same write at interior positions whenever the clamped feature’s activation is roughly constant across windows. Only the fourth, `txc_slab`, applies a temporally structured write to the residual stream. This note derives each claim explicitly. The organising fact is a decomposition: a decoder slab splits into a position-independent (DC) part and a mean-zero (AC) part, the DC part is exactly what a per-token dictionary can already express, and three of the four procedures write only the DC part.

1 Setup and notation

Fix a transformer, a hook layer ℓ , and a prompt of S tokens. Write $h \in \mathbb{R}^{S \times d}$ for the residual stream at layer ℓ , with $h_p \in \mathbb{R}^d$ the activation at token p . A *steering intervention* replaces h by $h + \Delta$ for some $\Delta \in \mathbb{R}^{S \times d}$. Everything below is a recipe for choosing Δ .

Let T be the window length and h (overloaded, but standard) the dictionary width. The two architectures differ in the shape of their decoder.

Per-token SAE. The code has one vector per token, $z \in \mathbb{R}^{S \times h}$, and the decoder is $W_{\text{dec}} \in \mathbb{R}^{h \times d}$:

$$\hat{x}_p = \sum_j z_{p,j} W_{\text{dec}}[j, :] + b_{\text{dec}}. \quad (1)$$

Latent j owns a single *direction* $W_{\text{dec}}[j, :] \in \mathbb{R}^d$.

Temporal crosscoder (TXC). The code is *shared across the window*, $z \in \mathbb{R}^h$ per window, and the decoder carries a position axis, $W_{\text{dec}} \in \mathbb{R}^{h \times T \times d}$:

$$\hat{x}_t = \sum_j z_j W_{\text{dec}}[j, t, :] + b_{\text{dec}}, \quad t = 0, \dots, T-1. \quad (2)$$

Latent j owns an entire matrix. We call it the *slab*:

$$P^{(j)} := W_{\text{dec}}[j] \in \mathbb{R}^{T \times d}, \quad P_t^{(j)} = W_{\text{dec}}[j, t, :] \in \mathbb{R}^d. \quad (3)$$

The row $P_t^{(j)}$ is the direction the latent writes at position t . Because the rows are free to differ, $\text{rank } P^{(j)}$ can exceed 1; that is the entire structural difference between the two architectures.

Definition 1 (DC/AC decomposition). For a slab $P \in \mathbb{R}^{T \times d}$ define the *position mean* $\bar{P} := \frac{1}{T} \sum_{t=0}^{T-1} P_t \in \mathbb{R}^d$ and split

$$P = \underbrace{\mathbf{1}_T \otimes \bar{P}}_{\text{DC part}} + \tilde{P}, \quad \tilde{P}_t := P_t - \bar{P}, \quad \sum_t \tilde{P}_t = 0. \quad (4)$$

The DC part is rank 1 and constant in t . The AC part is mean-zero across positions and carries all of the slab’s temporal structure.

Remark 2. The DC part is precisely what a per-token dictionary can produce: writing a fixed direction v at every position is $\mathbf{1}_T \otimes v$. So $\tilde{P}$ is the *only* component of a slab that a per-token dictionary cannot express at any steering strength. Any procedure that discards $\tilde{P}$ cannot, even in principle, exhibit an advantage for the crosscoder’s extra expressiveness.

2 The four procedures

2.1 V7 — tiled broadcast (the paper’s default)

Source: `_build_v7_hook` in `src/temp_bench/case_studies/steering.py`; selected by `protocol="v7"`, which is the default in `SteeringConfig` and in the `experiments/c5_steering/run.py` CLI.

Algorithm.

1. Tile the prefix into non-overlapping blocks of length T starting at token 0: block starts bT for $b = 0, \dots, \lfloor S/T \rfloor - 1$. If $S \bmod T > 0$, append one further window aligned to the *right* edge, starting at $S - T$.
2. Encode each block to get its code $z^{(b)}$.
3. **Clamp** coordinate j to an absolute value s : $z^{(b),c} = z^{(b)} + (s - z_j^{(b)})e_j$.
4. Decode both codes and take the difference, $\delta^{(b)} := \hat{x}(z^{(b),c}) - \hat{x}(z^{(b)}) \in \mathbb{R}^{T \times d}$.
5. **Average over the window**, $\bar{\delta}^{(b)} := \frac{1}{T} \sum_t \delta_t^{(b)} \in \mathbb{R}^d$.
6. Write $\bar{\delta}^{(b)}$ *identically* to all T positions of block b .

What it writes. By (2) the decoder is linear in z , so clamping a single coordinate gives

$$\delta_t^{(b)} = (s - z_j^{(b)}) P_t^{(j)} = c_b P_t^{(j)}, \quad c_b := s - z_j^{(b)} \in \mathbb{R}. \quad (5)$$

The clamp produces a scalar multiple of the slab — nothing more. Averaging and broadcasting:

$$\boxed{\Delta_p^{\text{V7}} = c_b \bar{P}^{(j)} \text{ for every } p \text{ in block } b.} \quad (6)$$

2.2 PP — per-position (the paper’s fallback)

Source: `_build_pp_hook`, selected by `protocol="pp"`. Used only when a V7 pre-test fails a coherence threshold.

Algorithm. Identical to V7 through step 4, except the windows slide with stride 1 ($K = S - T + 1$ of them), and instead of averaging over t the per-position deltas are scattered back to absolute positions and averaged over *overlap*:

$$\Delta_p^{\text{PP}} = \frac{1}{n_p} \sum_{w: 0 \leq p-w \leq T-1, 0 \leq w \leq K-1} c_w P_{p-w}^{(j)}, \quad n_p := \#\{w\}. \quad (7)$$

Window w covers absolute positions $w, \dots, w + T - 1$, so its row τ lands at $p = w + \tau$; equivalently row index $\tau = p - w$.

2.3 txc_slab — the sprint’s crosscoder arm

Source: `experiments/temporal_screen/txc_wins/harness.py`. The document is cut into T content-aligned segments and the raw decoder slab is injected once, normalised to unit Frobenius norm:

$$\Delta_t^{\text{slab}} = \alpha \frac{P_t^{(j)}}{\|P^{(j)}\|_F} \quad (8)$$

Row t is applied to segment t , exactly once, with no averaging.

2.4 txc_flat — the sprint’s profile-removal control

Same latent, same total injected norm, temporal profile deleted. With $P := P^{(j)} / \|P^{(j)}\|_F$ and $m := \frac{1}{T} \sum_t P_t$,

$$\Delta_t^{\text{flat}} = \alpha \frac{m}{\sqrt{T} \|m\|_2} \quad (\text{independent of } t). \quad (9)$$

Its purpose in the sprint is adversarial: if steering with Δ^{flat} matches steering with Δ^{slab} , the temporal profile is doing nothing and no temporal claim is available.

3 Equivalences

Proposition 3 (V7 is the flat control). *For a temporal crosscoder, Δ^{V7} and Δ^{flat} are the same intervention up to a positive scalar. Specifically, both are constant in position and both point along $\bar{P}^{(j)}$; they differ only in magnitude, c_b versus $\alpha / (\sqrt{T} \|m\|_2)$.*

Proof. Immediate from (6) and (9): the V7 write is $c_b \bar{P}^{(j)}$ and the flat write is a positive multiple of $m = \bar{P}^{(j)} / \|P^{(j)}\|_F$, which is a positive multiple of $\bar{P}^{(j)}$. Neither depends on t . $\square$

Proposition 4 (PP is a convolution, and collapses to flat in the interior). *Let p be an interior position, $T - 1 \leq p \leq K - 1$, so that all T windows containing p exist and $n_p = T$. Then*

$$\Delta_p^{\text{PP}} = \frac{1}{T} \sum_{\tau=0}^{T-1} c_{p-\tau} P_{\tau}^{(j)} = \frac{1}{T} (c \star P^{(j)})_p, \quad (10)$$

a cross-correlation of the scalar sequence $(c_w)_w$ with the rows of the slab. Writing $c_w = \bar{c} + \epsilon_w$ with $\sum \epsilon$ over the relevant window zero-mean,

$$\Delta_p^{\text{PP}} = \underbrace{\bar{c} \bar{P}^{(j)}}_{\text{the V7 / flat write}} + \frac{1}{T} \sum_{\tau} \epsilon_{p-\tau} \tilde{P}_{\tau}^{(j)}. \quad (11)$$

In particular, if the clamped feature’s activation is constant across windows ($\epsilon \equiv 0$), then $\Delta_p^{\text{PP}} = \bar{c} \bar{P}^{(j)}$ exactly — identical to V7.

Proof. For interior p , as w runs over $p - T + 1, \dots, p$ the row index $\tau = p - w$ runs over $T - 1, \dots, 0$, hitting each row exactly once; substituting into (7) with $n_p = T$ gives the first line. For the second, substitute $c_{p-\tau} = \bar{c} + \epsilon_{p-\tau}$ and use $\frac{1}{T} \sum_{\tau} P_{\tau} = \bar{P}$ together with $P_{\tau} = \bar{P} + \tilde{P}_{\tau}$ and $\sum_{\tau} \tilde{P}_{\tau} = 0$. $\square$

Corollary 5. *PP recovers temporal structure only through the variation of the clamped latent’s activation across windows, and only at positions where that variation is appreciable relative to its mean. At the sequence edges ($p < T - 1$ or $p > K - 1$) fewer than T windows contribute, $n_p < T$, and a partial profile survives for a boundary layer of width $T - 1$.*

Proposition 6 (Only the slab writes AC). *Using (4),*

$$\Delta_t^{\text{slab}} = \frac{\alpha}{\|P^{(j)}\|_F} \left(\bar{P}^{(j)} + \tilde{P}_t^{(j)} \right),$$

whereas Δ^{V7} and Δ^{flat} are exactly (scalar) $\cdot \bar{P}^{(j)}$ and contain no $\tilde{P}$ component, and Δ^{PP} contains only the ϵ -modulated image of $\tilde{P}$ in (11).

4 Why this is the whole question

A per-token dictionary latent, steered arbitrarily, reaches writes of the form $\mathbf{1}_T \otimes v$ — one direction, possibly with a time-varying gain if the coefficient is allowed to vary, hence any rank-1 matrix. The crosscoder’s claim to extra power rests entirely on $\tilde{P}$: the part of the slab that is *not* expressible as a constant direction.

Consequently:

A steering protocol that averages over the window before writing cannot demonstrate a crosscoder advantage in expressiveness, because it deletes the only component in which such an advantage could appear.

This is not a claim that the protocol is wrong — it is even-handed, since for a per-token SAE ($z \in \mathbb{R}^{S \times h}$) V7 averages the per-position delta too. It is a claim about what the resulting numbers can and cannot support.

4.1 A worked micro-example

Take $T = 3$, $d = 2$, and a latent whose slab writes $+e_1$ early, nothing in the middle, and $-e_1$ late, plus a constant nudge along e_2 :

$$P = \begin{pmatrix} 1 & \frac{1}{3} \\ 0 & \frac{1}{3} \\ -1 & \frac{1}{3} \end{pmatrix}, \quad \bar{P} = \left(0, \frac{1}{3} \right), \quad \tilde{P} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ -1 & 0 \end{pmatrix}. \quad (12)$$

The slab has rank 2. Its temporally interesting content — push one way at the start, the other way at the end — lives entirely in the first coordinate and is *exactly mean-zero*. Therefore:

- Δ^{slab} writes $+e_1$ at $t = 0$ and $-e_1$ at $t = 2$: the effect is present.

- $\Delta^{V7} = \Delta^{\text{flat}} \propto (0, \frac{1}{3})$ at every position: the e_1 structure has cancelled *identically*. The intervention is blind to it.
- Δ^{PP} at an interior p equals $\frac{1}{3}(c_p - c_{p-2})e_1 + \frac{1}{3}\bar{c}e_2$, so it sees the e_1 structure only to the extent that the latent’s activation differs between windows two apart.

An antisymmetric temporal profile — arguably the canonical “temporal” feature, and the shape of any “first versus last” or “before versus after” effect — is in the exact kernel of the tiled-broadcast protocol.

5 Two axes of difference beyond the profile

Position indexing: tokens versus content. V7 tiles from token 0: block b occupies absolute tokens $bT, \dots, bT + T - 1$, so the row index of a given piece of text is $p \bmod T$, which depends on the arbitrary phase of the tiling relative to the prompt. Prepending a single token permutes every row assignment. PP inherits the same token-indexing, softened by the overlap average. By contrast `txc_slab` and `txc_flat` segment the *document*, so “position t ” means “segment t of this example” in every draw. Even a protocol that preserved $\tilde{P}$ would apply it at a random phase under token tiling, which for content-aligned features averages it toward zero across examples.

Dose semantics: activation space versus residual space. V7 and PP fix the *latent value* $z_j = s$ (“absolute clamp”), and let the decoder norm determine how hard the residual stream is pushed; by (5) the injected norm is $|s - z_j| \|P^{(j)}\|_F$, which varies per document and, critically, across architectures whose decoder norms differ. `txc_slab` and `txc_flat` fix the *injected Frobenius norm* to α and let the implied latent value fall where it may.

Neither convention makes the comparison automatically fair. Matching on latent value leaves the injected norm free; matching on injected norm leaves the per-position norm free, and a concentrated write then places $\sqrt{T}$ times more norm on its dominant position than a broadcast write does, so it leaves the linear regime at a lower nominal dose.¹ A single scalar dose is therefore not comparable across procedures, and dose–response curves should be reported instead.

6 Summary

	V7 tiled bcast	PP per-position	txc_slab	txc_flat
windowing	non-overlap T -blocks	stride-1, overlap-avg	one aligned window	one aligned window
write at pos. p	$c_b \bar{P}$	$\frac{1}{T} \sum_{\tau} c_{p-\tau} P_{\tau}$	$\alpha P_t / \ P\ _F$	$\alpha m / (\sqrt{T} \ m\)$
DC part	yes	yes	yes	yes
AC part	no	only via $\text{var}(c_w)$; none if c const	yes, exactly	no
rank of write	1	1 in interior	up to $\min(T, d)$	1
position index	token, phase-arbitrary	token, phase-arbitrary	content segment	content segment
dose fixed in	activation ($z_j = s$)	activation ($z_j = s$)	residual ($\ \Delta\ _F$)	residual

The three-line version. V7 and `txc_flat` are the same intervention, one published as the method and one built as the null. PP is a convolution of the slab with the latent’s activation profile, and

¹Measured on the StruQ cell in this sprint: the gradient slab, whose mass sits almost entirely on one segment of twelve, beat the best constant write at $\alpha \in \{0.02, 0.05, 0.10, 0.15, 0.25\}$ and lost to it at $\alpha \in \{0.5, 1.0\}$, a clean saturation crossover. See `results/txc_wins/geometrystruqposx.lowdose.json`.

reduces to the same constant write wherever that activation is flat. `txc_slab` is the only one of the four that delivers $\tilde{P}$ to the residual stream.

7 The AC part is not a technicality: a measured case

The decomposition above is algebra; whether $\tilde{P}$ carries any usable effect is empirical. On the one task in this sprint where a crosscoder cleanly beats every learned per-token arm — the Lost-in-the-Middle retrieved-document-position cell, gold at 0 versus gold at 4 — it carries essentially *all* of it. Median over three dictionary initialisations, Qwen2.5-1.5B-Instruct layer 14, $T = 11$, held-out content, every write at identical injected Frobenius norm (`results/txc_wins/litm_0v4.tr_tsaep.ds{0,1,2}.json`):

arm	matched $ \alpha = 0.5$	peak over grid
<code>txc_slab</code> (writes DC + AC)	0.843	2.493
<code>tsaep_broadcast</code> (published T-SAE)	0.238	0.257
<code>sae_broadcast</code> (TopK SAE)	0.124	1.474
<code>tsae_broadcast</code> (attention tSAE)	0.100	0.118
<code>txc_flat</code> (DC only $\equiv$ V7)	0.044	0.180
<code>random_slab</code> (null)	0.048	0.182

`txc_flat` is the V7 write up to a scalar (Proposition 3), and it measures 0.044 against a *random* slab’s 0.048 — indistinguishable, at matched dose and again at peak (0.180 versus 0.182). The same latent, writing its full slab, measures 0.843: a factor of roughly 19.

On this cell the paper’s steering protocol would have scored the crosscoder at the random-write level. The DC component of the winning latent is worth nothing; the entire effect lives in $\tilde{P}$, which V7 averages away.

This is one cell, one model and one metric, and it does not establish that the same holds on the paper’s tasks. What it does establish is that the gap between the protocols is not a small correction: a latent can be genuinely useful and completely invisible to a window-averaging intervention. Any negative steering result obtained under V7 is therefore uninformative about whether the latent could have steered — which is a different and much weaker statement than those results are usually read as making.

Provenance and what is not established here

The V7 and PP descriptions are read from `origin/temp-bench-anon:src/temp_bench/case_studies/steering.` and the protocol default from `SteeringConfig` (`protocol: str = "v7"`) and `experiments/c5_steering/run.py` (`--protocol, default="v7"`), corroborated by `experiments/c5_steering/README.md` (“V7-tile-broadcast” pipeline). *The manuscript itself has not been checked.* If a published figure was produced with `--protocol pp`, that figure belongs in the second column rather than the first.

Propositions 3 and 4 are algebraic consequences of decoder linearity and hold exactly. What is *not* established here is empirical: how large $\text{var}(c_w)$ is in practice, and therefore how far PP sits from flat on real prompts; and whether the $\tilde{P}$ component carries useful effect on the paper’s tasks, which requires running `txc_slab`-style writes on them rather than reasoning about the protocol.